

# Writing Small Numbers in Scientific Notation

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## MA.8.NSO.1.4

*I can write very small numbers in scientific notation, read them back, and find how many times larger or smaller one number is than another.*

*Yesterday the exponents on the 10 were positive.*

**Today they are negative — and the numbers get very small.**

# Warm-Up

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*On your own. Four minutes. No notes.*

1. Write 4,300,000 in scientific notation.
2. Write  $10^{-3}$  as a fraction.
3.  $\sqrt{95}$  is between which two whole numbers?
4. Write 0.007 as a fraction.

# Warm-Up

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*Answers.*

1. Write 4,300,000 in scientific notation.  **$4.3 \times 10^6$**
2. Write  $10^{-3}$  as a fraction.  **$\frac{1}{1,000}$**
3.  $\sqrt{95}$  is between which two whole numbers? **9 and 10**
4. Write 0.007 as a fraction.  **$\frac{7}{1,000}$**

# Notes

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*Copy the table.*

## I. Negative powers of ten name the decimal places

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| $10^0$ | $10^{-1}$ | $10^{-2}$ | $10^{-3}$ | $10^{-4}$ | $10^{-5}$ |
|--------|-----------|-----------|-----------|-----------|-----------|
| 1      | 0.1       | 0.01      | 0.001     | 0.0001    | 0.00001   |

$$10^{-n} = \frac{1}{10^n}$$

$$10^{-3} = \frac{1}{1,000} = 0.001$$

**A negative exponent puts the 1 in that decimal place:  $10^{-4}$  is a 1 in the fourth place.**

# Notes

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## II. Small numbers in scientific notation

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$$0.000007 = 7 \times 10^{-6}$$

$$0.0000672 = 6.72 \times 10^{-5}$$

$$3.4 \times 10^{-9} = 0.0000000034$$

**The coefficient is still at least 1 and less than 10. The exponent is negative for a number less than 1.**

The exponent is how many places the point moves:  $0.0000672 \rightarrow 6.72$  is five places right, so  $10^{-5}$ .

# Notes

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## III. How many times larger — or smaller?

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$$\frac{7.2 \times 10^{-4}}{1.8 \times 10^{-7}} = 4 \times 10^3 = 4,000$$

$$\frac{3 \times 10^8}{3 \times 10^{-4}} = 10^{12}$$

**Divide the coefficients. Divide the powers of 10 — subtract the exponents. Multiply the results.**

Which is bigger,  $10^{-5}$  or  $10^{-8}$ ? The one with the larger exponent:  $-5 > -8$ , so  $10^{-5}$ .

# Example 1

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*A quantity in words, to scientific notation.*

The density of oxygen is 1,332 millionths of a gram per cubic centimeter.

Write this value as a decimal, then in scientific notation.

# Example 1

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*Worked.*

$$\frac{1,332}{1,000,000} = 0.001332 \quad \text{1,332 millionths, as a decimal}$$

$$0.001332 = 1.332 \times 10^{-3} \quad \text{the 1 is in the thousandths place}$$

**Answer:  $0.001332 = 1.332 \times 10^{-3}$**

# Your Turn

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*Same steps, your numbers. Boards up when done.*

A human hair is about 67 hundred-thousandths of an inch thick.

Write this value in scientific notation.

# Your Turn

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*Answer.*

A human hair is about 67 hundred-thousandths of an inch thick.

Write this value in scientific notation.

0.00067 — the 6 is in the ten-thousandths place

**Answer:  $6.7 \times 10^{-4}$**

## Example 2

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*How many times larger?*

How many times larger is  $7.2 \times 10^{-4}$  than  $1.8 \times 10^{-7}$  ?

# Example 2

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*Worked.*

$$\frac{7.2}{1.8} = 4 \quad \text{coefficients}$$

$$\frac{10^{-4}}{10^{-7}} = 10^{-4 - (-7)} = 10^3 \quad \text{powers of 10: } -4 - (-7) = 3$$

$$4 \times 10^3 = 4,000 \quad \text{multiply the two results}$$

**Answer: 4,000 times larger**

# Your Turn

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*Same steps, your numbers. Boards up when done.*

How many times larger is  $5.2 \times 10^{-3}$  than  $2.6 \times 10^{-8}$  ?

# Your Turn

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*Answer.*

How many times larger is  $5.2 \times 10^{-3}$  than  $2.6 \times 10^{-8}$  ?

$2 \times 10^5$

**Answer: 200,000 times larger**

# Whiteboards • Question 1 of 9

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*Boards up on three.*

$$10^{-4}$$

**Answer it.**

*Write it as a decimal.*

# Whiteboards • Question 1 of 9

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*Answer.*

$$10^{-4}$$

a 1 in the fourth decimal place

**Answer: 0.0001**

# Whiteboards • Question 2 of 9

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*Boards up on three.*

Write 0.003 in scientific notation.

**Answer it.**

*Which place is the 3 in?*

# Whiteboards · Question 2 of 9

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*Answer.*

Write 0.003 in scientific notation.

3 thousandths

**Answer:  $3 \times 10^{-3}$**

# Whiteboards • Question 3 of 9

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*Boards up on three.*

Write  $2.5 \times 10^{-4}$  in standard form.

**Answer it.**

*Move the point four places left.*

# Whiteboards • Question 3 of 9

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*Answer.*

Write  $2.5 \times 10^{-4}$  in standard form.

the 2 lands in the ten-thousandths place

**Answer: 0.00025**

# Whiteboards • Question 4 of 9

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*Boards up on three.*

A machine's cutting tolerance is 0.0000988 inch.

The part it cuts is 4 inches long.

Write the tolerance in scientific notation.

**Answer it.**

*Every nonzero digit stays in the coefficient.*

# Whiteboards • Question 4 of 9

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*Answer.*

A machine's cutting tolerance is 0.0000988 inch.

The part it cuts is 4 inches long.

Write the tolerance in scientific notation.

the 9 is in the hundred-thousandths place

**Answer:  $9.88 \times 10^{-5}$**

# Whiteboards • Question 5 of 9

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*Boards up on three.*

Which is 0.00081 written in scientific notation?

**A.**  $8.1 \times 10^{-4}$

**B.**  $8.1 \times 10^{-3}$

**C.**  $81 \times 10^{-5}$

**D.**  $8.1 \times 10^4$

**Answer it.**

# Whiteboards • Question 5 of 9

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*Answer.*

Which is 0.00081 written in scientific notation?

**A.  $8.1 \times 10^{-4}$**

**B.  $8.1 \times 10^{-3}$**

**C.  $81 \times 10^{-5}$**

**D.  $8.1 \times 10^4$**

**Answer: A —  $8.1 \times 10^{-4}$**

# Whiteboards • Question 6 of 9

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*Boards up on three.*

How many times larger is  $6 \times 10^{-3}$  than  $2 \times 10^{-5}$  ?

**A.** 300

**B.** 3

**C.** 30

**D.** 0.03

**Answer it.**

# Whiteboards • Question 6 of 9

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*Answer.*

How many times larger is  $6 \times 10^{-3}$  than  $2 \times 10^{-5}$  ?

**A. 300**

**B. 3**

**C. 30**

**D. 0.03**

**Answer: A — 300**

# Whiteboards • Question 7 of 9

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*Boards up on three.*

Which is greater,  $3.9 \times 10^{-5}$  or  $7.4 \times 10^{-8}$  ?

**Answer it.**

*Compare the exponents first.*

# Whiteboards • Question 7 of 9

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*Answer.*

Which is greater,  $3.9 \times 10^{-5}$  or  $7.4 \times 10^{-8}$  ?

$-5 > -8$ , so  $10^{-5}$  is the larger power

**Answer:  $3.9 \times 10^{-5}$**

# Whiteboards • Question 8 of 9

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*Boards up on three.*

A computer displays a number as  $4.2\text{E}-6$ .

Write the number in standard form.

**Answer it.**

*E-6 means  $\times 10^{-6}$ .*

# Whiteboards • Question 8 of 9

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*Answer.*

A computer displays a number as  $4.2\text{E}-6$ .

Write the number in standard form.

$$4.2 \times 10^{-6}$$

**Answer: 0.0000042**

# Whiteboards • Question 9 of 9

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*Take your time. Boards up when you have written it.*

Fishing line A is  $2 \times 10^{-2}$  inch thick.

Fishing line B is  $6 \times 10^{-2}$  inch thick.

Which is thicker, and how many times? Name a thickness between them.

**Write your answer in sentences.**

*Same power of 10 — compare the coefficients.*

# Whiteboards · Question 9 of 9

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*Answer.*

Fishing line A is  $2 \times 10^{-2}$  inch thick.

Fishing line B is  $6 \times 10^{-2}$  inch thick.

Which is thicker, and how many times? Name a thickness between them.

0.02 in and 0.06 in

**Answer: B; 3 times; e.g.  $4 \times 10^{-2}$  in.**

# Independent Set

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*Six questions. On your own, in silence.*

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

# IXL

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*Last five minutes.*

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

## **Today's skills — all required**

Convert between standard and scientific notation: small numbers — 85M

Compare small numbers — KDR

Also consider: Powers of ten with negative exponents — SKD